

Kevin Luo

Chandler, AZ | kevinluo423@gmail.com | (480) 335-0826 | linkedin.com/in/kevin-luo7 | US Citizen

Education

Georgia Institute of Technology (GT) | Atlanta, GA
Bachelor of Science in Computer Engineering, GPA 4.00

Expected Graduation, May 2028

Work Experience

Intel | Chandler, AZ

May 2026 – August 2025

Incoming SoC Design Verification Intern

- Will develop UVM testbenches and Python automation flows for IPs in the Networking and Communications Solutions unit

Microchip | Chandler, AZ

June 2025 – July 2025

Reliability Engineering Intern

- Developed Python Retrieval-Augmented Generation (RAG) pipeline with Docling, HuggingFace, and Streamlit UI for scalable AI internal document search and reliability analysis system to reduce department information retrieval time by ~97%
- Built and programmed C++ firmware for a low-power 433 MHz RF wireless temperature logger prototype featuring custom-soldered shield and custom 3D-printed enclosure for burn-in oven and test environment validation
- Processed Engineering Change Notices (ECNs) and revisions for burn-in schematic and pinout revisions for consistency among company-wide technical standards

Project Experience

SiliconJackets | Digital Verification Member

August 2025 – Present

- Implemented SystemVerilog RTL for 64-bit calculator with dual 32-bit adders, 5-state FSM control, and SRAM interface, and verified its functionality with custom testbenches for 99% code coverage via Cadence SimVision and Synopsys Verdi
- Reached 100% functional coverage for verification of 6-port (4r/2w) register file via SystemVerilog UVM testbench, backed by architectural specs, test plans, SVA, Cadence Xcelium, and Cadence IMC in preparation for tapeout of a RISC-V core

HyTech FSAE Electric Racing | Software Engineer & Firmware Member

August 2025 – Present

- Developing C++ Teensy4.1 firmware with PlatformIO for ADC data acquisition on the vehicle's main front and accumulator control systems with Python scripting and oscilloscope signal validation of SPI and UDP Ethernet communications
- Integrating VectorNav GNSS/INS drivers into Raspberry Pi 'DriveBrain' with Nix and Docker build systems for EV state-estimation
- Designed and soldered 5Hz flashing PCB keychain on Altium using an astable 555-Timer and USB-C power input

High-Security PYNQ-Z1 SoC Transaction Machine | Digital Design & Verification

January 2026 – Present

- Learning Questa Formal for SVA formal verification of shared-memory interface in dual-core ARM Cortex and PicoRV32 system
- Designing and integrating Physical Memory Attribution (PMA) with custom Verilog testbenches in Vivado for a custom RISC-V Privileged Architecture implementation to enforce memory access policies and enhance overall system reliability

FIRST Robotics Competition Team 9704 | Founder & Team Captain

September 2023 – May 2025

- Founded and lead school's 30+ member robotics team, raising \$25,000 in funding and winning 5 competition awards
- Constructed robotic control systems via CAD, mechanical assembly, and integration of actuators, sensors, and microcontrollers
- Produced 200+ pages of comprehensive business and technical documentation for training purposes and knowledge retention

Relevant Coursework

Digital Design Lab (Ongoing): Hands-on FPGA implementation via Quartus Prime; VHDL design; timing analysis & signal debugging

Programming HW/SW Systems (Ongoing): RISC-V ISA Assembly; embedded C; Computer Architecture; Linux; networking

Data Structures & Algorithms (Ongoing): Linear and hierarchical data structures; sorting and graph algorithms; asymptotic analysis

Digital System Design: Combinational & sequential logic; digital circuit analysis; boolean algebra; FSM design, datapath architecture

Skills

Programming & HDL: SystemVerilog, C++, C, Java, Python, Verilog, VHDL, TCL (Basic), Cmake, Nix (Basic), RISC-V Assembly

Tools & Platforms: Altium, Cadence, Quartus, Synopsys, Linux (Ubuntu), Github, FPGA, Oscilloscope, WireShark, Protobuf, Vivado

Competencies: SPI, I2C, CAN, Universal Verification Methodology (UVM), POSIX Ethernet, Digital Circuit Analysis, Neural Networks